

PENDLETON EASTERN OREGON, OR, USA

WMO: 726880

Lat: 45.698N

Lon: 118.855W

Elev: 453

StdP: 96.00

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.0	-9.1	-16.1	1.0	-12.1	-12.5	1.3	-7.6	14.8	7.0	12.9	7.8	2.1	150	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	36.2	18.2	34.3	17.6	32.4	17.0	19.5	33.0	18.5	31.9	17.7	30.4	4.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	11.0	21.9	13.5	10.2	21.6	12.4	9.5	21.4	56.9	32.8	53.9	31.8	51.1	30.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.8	9.1	DB	-15.3	39.7	5.0	1.8	-18.9	41.0	-21.8	42.1	-24.7	43.1	-28.3	44.4	(4)
(5)			WB	-15.7	21.1	4.8	1.8	-19.2	22.4	-22.0	23.5	-24.7	24.5	-28.3	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	1.9	3.5	7.1	10.2	14.5	18.3	23.2	22.5	17.8	10.9	5.2	1.3		
	DBStd	8.56	5.34	4.60	3.59	3.32	3.84	3.69	3.40	3.28	3.68	3.75	4.80	5.26		
	HDD10.0	1040	254	186	101	37	5	0	0	0	1	32	153	272		
	HDD18.3	2907	511	416	347	245	130	44	3	4	53	231	394	528		
	CDD10.0	1555	2	3	12	42	145	249	410	386	234	60	9	2		
	CDD18.3	382	0	0	0	1	12	43	155	132	37	1	0	0		
	CDH23.3	5529	0	0	1	17	219	684	2238	1805	541	24	1	0		
	CDH26.7	2639	0	0	0	2	66	276	1196	896	199	5	0	0		
(14) Wind	WSAvg	3.7	3.1	3.4	4.0	4.3	4.1	4.2	3.9	3.8	3.6	3.5	3.4	3.3		
(15) Precipitation	PrecAvg	315	38	28	32	29	30	22	8	11	14	25	38	40		
	PrecMax	474	100	76	72	71	81	57	37	66	56	68	96	119		
	PrecMin	172	10	2	7	2	1	0	0	0	0	0	5	5		
	PrecStd	61	21	16	15	17	20	14	8	15	14	15	17	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	16.8	20.8	25.4	31.4	36.0	39.1	38.0	34.3	26.5	19.0	16.0		
		MCWB	9.8	10.0	11.3	14.6	16.9	18.8	19.5	18.8	17.5	14.7	12.4	9.8		
	2%	DB	12.5	13.7	17.8	22.6	28.5	32.5	36.7	35.7	31.2	22.6	16.2	13.0		
		MCWB	8.0	8.6	10.3	13.3	16.0	17.6	18.4	17.9	16.4	13.2	10.7	8.4		
	5%	DB	10.5	11.6	15.6	19.8	25.8	29.8	35.1	33.8	29.0	20.0	13.9	10.6		
		MCWB	6.9	7.4	9.3	11.8	14.8	16.2	17.7	17.3	15.6	12.0	9.4	7.0		
	10%	DB	8.8	9.9	13.7	17.2	23.0	27.3	33.0	31.7	26.6	17.6	11.8	8.3		
		MCWB	5.6	6.2	8.3	10.3	13.7	15.2	17.2	16.7	14.6	10.9	7.9	5.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	11.0	12.7	16.1	18.5	19.8	21.0	20.6	19.0	16.1	13.9	10.3		
		MCDB	13.8	15.0	18.4	23.4	28.8	33.4	35.8	34.5	31.0	23.7	18.0	14.6		
	2%	WB	8.5	9.3	11.1	13.9	16.9	18.2	19.7	19.1	17.5	14.1	11.6	8.6		
		MCDB	11.8	13.0	16.1	21.4	26.3	30.7	33.6	32.7	28.5	21.0	15.2	12.1		
	5%	WB	7.1	7.7	10.0	12.4	15.6	17.0	18.7	18.2	16.3	12.8	9.9	7.2		
		MCDB	10.1	11.2	14.7	18.9	23.9	28.1	32.9	31.4	26.9	18.6	13.1	10.5		
	10%	WB	5.8	6.5	8.8	10.9	14.4	15.8	17.7	17.3	15.2	11.6	8.4	5.6		
		MCDB	8.5	9.7	13.3	16.3	21.8	25.4	31.2	30.0	24.9	16.7	11.4	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	8.3	10.7	11.9	13.0	14.2	16.9	16.3	14.8	12.0	8.3	6.5		
		MCDBR	9.3	10.2	13.6	16.0	17.5	18.5	19.4	18.9	18.2	15.3	10.4	9.0		
		MCWBR	6.1	6.4	7.7	8.4	7.6	6.9	7.3	7.4	7.8	7.9	6.3	5.8		
		MCDBR	9.1	9.7	12.6	14.9	15.6	17.0	17.9	17.2	16.6	13.7	9.6	8.6		
	5% WB	MCWBR	6.3	6.3	7.5	8.2	7.3	6.8	7.2	7.2	7.4	7.5	6.3	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.290	0.304	0.336	0.355	0.353	0.353	0.353	0.338	0.312	0.293	0.285		
	taud		2.539	2.555	2.533	2.425	2.393	2.424	2.430	2.421	2.445	2.522	2.537	2.523		
	Ebn at Noon		837	906	935	924	912	912	907	895	881	861	820	798		
	Edh at Noon		64	75	89	108	115	112	110	107	96	77	63	58		
(44) All-Sky Solar Radiation	RadAvg		1.27	2.25	3.52	5.17	6.27	7.17	7.71	6.51	4.81	2.97	1.58	1.03		
	RadStd		0.14	0.27	0.28	0.33	0.47	0.37	0.26	0.36	0.29	0.22	0.13	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.55	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A

Nomenclature: See separate page